

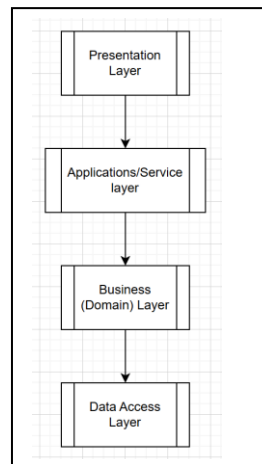
(1) Software Architecture Style -

Layered Architecture

Since, the software system is a generic automated essay evaluation platform, with well - defined functional requirements and a centralized deployment model. It does not require a series of distributed services, inter layer communications, thus choosing the layered architecture will provide us proficient separation of concerns into key modules and classes.

The system is divided into four layers -

- (i) Presentation layer
- (ii) Application layer
- (iii) Business layer
- (iv) Data Access layer



Granularity of components in each layer -

- Presentation layer – Includes User interfaces (like, Login ui, essay submission ui, feedback display, etc).
- Application layer/Submission layer – Includes modules that provide services to the system (like, Authentication service, essay submission service, essay evaluation service, etc.).
- Business layer/domain layer - Includes the business rules and the main(core) logic components, like User, RegisteredUser, Student, Teacher, EvalModule, Essay, Organization.
- Data Access layer – Acts as the data storage unit for the system, the database is isolated and protected from unauthorized access (like, User database, Essay database, etc.).

Here, each layer has a distinct responsibility with each having smaller classes and functions. Dependencies flow top -> down only,

i.e. higher layer components depend on lower layers but not vice versa. This architecture provides an adequate mixture of fine grained and coarse - grained components, as the separation of group of modules into layers provides flexible operations, without communication overheads, as there are not distributed microservices.

Why we chose Layered Architecture -

- Scalability - The system can be scaled adequately for moderate number of users, each module can be optimized independent of the other layers, as they fall in the domain layer, and there is no need for distributed services in this software.
- Maintainability – There exists clear separation of concerns, i.e. any error in presentation layer will not affect the business layer, and the database changes is isolated in a separate layer.

- Performance - The single deployable design allows removing communication overhead, which could occur in a microservices architecture. Hence, there is no network latency between the layers.
- Simplicity - The layered architecture removes complex implementation, with the overall system being packed into a single deployable unit, at the end of the development phase.

(2) Components present in this software are -

(i)System UI (Presentation layer) components -

- Login interface
- New user registration interface
- Essay submission
- feedback display
- user dashboard
- Essay analytics.

(ii)Application/Service layer components –

- Authentication service
- user management service
- essay evaluation service
- essay management service
- Organization management service.

(iii)Business (Domain) layer components -

- User
- RegisteredUser
- UnregisteredUser
- Student
- Teacher
- Administrator
- Essay
- EvalModule
- Organization.

(iv)Data Access layer -

- User database
- Essay database
- Feedback database
- Database connector.